Q U A N T U M L A T T I C E M O D E L

Planck Energy Density Cap and Minimal Saturated-Core Completion

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Abstract

The Quantum Lattice Model (QLM) formulates physics as deterministic phase–action evolution on a discrete Planck-scale spacetime lattice. The foundational QLM primitive is the covariant phase–flow law $E = \hbar\, d\theta/d\tau$, so that a saturated proper-time tick ($\Delta\theta = 1$, $\Delta\tau = t_P$) transports one quantum of reduced action $\hbar$ per Planck four–volume cell $\mathcal{V}_4 = \ell_P^3 t_P$. This implies a universal local Planck energy density scale $u_P = \hbar/(\ell_P^3 t_P)$, which in this work is promoted to an explicit local constraint: a Planck energy-density cap on coherent phase–action throughput, $u(r) \leq u_P$.

We show that this cap implies a maximum lattice mass density and enforces a minimal, non-singular interior completion for gravitational collapse. The resulting interior consists of a finite saturated core whose radius scales as the cube root of the total mass, while the exterior gravitational field remains exactly Schwarzschild, depending only on the total enclosed mass. No exotic matter, additional parameters, or modifications of General Relativity outside the core are introduced. Numerical examples spanning terrestrial, stellar, and supermassive mass scales confirm the analytic scaling relations. Singular collapse is thereby forbidden as a direct consequence of local saturation of the QLM density bound.

1 Introduction

Classical General Relativity predicts the formation of spacetime singularities under gravitational collapse, signaling a breakdown of the theory at high densities. While such singularities are widely interpreted as indicators of missing ultraviolet physics, many proposed resolutions introduce additional fields, exotic states of matter, or phenomenological modifications of the Einstein equations. A central open question is whether singular collapse can be avoided as a direct and unavoidable consequence of a more fundamental description of spacetime itself, without altering the exterior gravitational field or introducing free parameters.

The Quantum Lattice Model (QLM), developed in *Quantum Lattice Model: Reduced-Action Foundations & Planck-Unit Derivations* [12], formulates physics as deterministic

phase–action evolution governed by the covariant primitive

$$E = \hbar \frac{d\theta}{d\tau}, \qquad c = \frac{\ell_P}{t_P}.$$

In this framework, a saturated proper-time tick ($\Delta\theta = 1$, $\Delta\tau = t_P$) transports one quantum of reduced action $\hbar$ per Planck four–volume cell

$$\mathcal{V}_4 = \ell_P^3 t_P.$$

From this primitive structure follows a universal local Planck energy density,

$$u_P = \frac{\hbar}{\ell_P^3 t_P},$$

expressed entirely in terms of the lattice primitives $\{\hbar, \ell_P, t_P\}$. This density was derived in the foundational work as a direct consequence of the covariant phase–flow rule, rather than as an imposed cutoff or phenomenological bound.

The present paper makes explicit a consequence already implicit in the QLM foundations: the Planck energy density functions as a local, action-limited constraint on coherent phase–action throughput. When treated as a strict local bound,

$$u(r) \leq u_P,$$

this constraint implies a maximum lattice mass density and forbids unlimited compression. Gravitational collapse must therefore terminate in a finite, non-singular interior configuration corresponding to a fully saturated lattice state.

The resulting interior completion consists of a compact saturated core whose radius scales as the cube root of the total mass. Importantly, the exterior gravitational field depends only on the enclosed mass and remains exactly Schwarzschild. No exotic matter, additional parameters, or modifications of General Relativity outside the core are required. The construction is fixed entirely by the QLM lattice primitives and their associated local density bound.

This paper is organized as follows. Section 2 reviews the QLM primitives and the gravitational impedance formulation. Section 3 formulates the Planck energy density cap and its interpretation as a local action-limited constraint. Section 4 derives the corresponding maximal mass density and the resulting saturated-core interior completion. Section 5 examines the scaling of the core radius relative to the Schwarzschild radius. Section 6 provides numerical examples spanning terrestrial, stellar, and supermassive mass scales, confirming the analytic scaling relations. Section 7 reinterprets black-hole thermodynamic quantities within the QLM impedance framework. Section 8 explores derived near-horizon consequences of the Planck density cap, including bandwidth limitations and echo scaling. Appendices provide numerical constants, derived Planck quantities, and a lattice-impedance formulation of gravitational radiation.

2 Definitions and QLM Primitives

We work in Quantum Lattice Model (QLM) primitives:

$$\ell_P, \quad t_P, \quad \hbar. \tag{2.1}$$

where ℓ_P is the Planck cell edge length, t_P is the Planck proper-time tick, and $\hbar$ is the reduced action quantum.

The primitive covariant phase–flow rule is

$$E = \hbar \frac{d\theta}{d\tau}, \tag{2.2}$$

so that a saturated tick $\Delta\theta = 1,\ \Delta\tau = t_P$ transports one quantum of reduced action $\hbar$.

The single spatial Planck cell volume is

$$\mathcal{V}_P = \ell_P^3, \tag{2.3}$$

and the corresponding Planck four–volume cell is

$$\mathcal{V}_4 = \ell_P^3 t_P. \tag{2.4}$$

QLM adopts the exact lattice kinematic identity

$$c \equiv \frac{\ell_P}{t_P}. \tag{2.5}$$

The primitive per–radian Planck energy is

$$E_P \equiv \frac{\hbar}{t_P}. \tag{2.6}$$

The QLM Planck mass follows directly as

$$m_P \equiv \frac{E_P}{c^2} = \frac{\hbar\, t_P}{\ell_P^2}. \tag{2.7}$$

All subsequent quantities in this work are expressed in terms of the invariant QLM primitives $\{\hbar, \ell_P, t_P\}$ and derived combinations thereof.

2.1 Gravitational Impedance / Throttle Factor

In the Quantum Lattice Model, gravity does not modify the local phase–action primitive $E = \hbar\, d\theta/d\tau$. Local proper-time evolution remains invariant. Instead, gravity alters the mapping between local proper time τ and the coordinate time measured by an asymptotic observer, as described by the Schwarzschild solution of General Relativity [2, 1, 3].

Gravitational time dilation is therefore interpreted as an effective transport impedance that throttles the rate at which local phase–action transport is observed relative to infinity.

For a spherically symmetric mass M, the gravitational impedance (redshift throttle) factor is

$$Z_g(r) = \left(1 - \frac{r_s}{r}\right)^{-1/2} = \left(1 - \frac{2GM}{c^2 r}\right)^{-1/2}, \tag{2.8}$$

which is the inverse of the Schwarzschild lapse factor.

Using the Planck identities $G = \ell_P^5/(t_P^3\hbar)$, $c = \ell_P/t_P$, and $m_P = \hbar t_P/\ell_P^2$, this becomes

$$Z_g(r) = \left[1 - 2\left(\frac{M}{m_P}\right)\left(\frac{\ell_P}{r}\right)\right]^{-1/2}. \tag{2.9}$$

The Schwarzschild radius is therefore

$$r_s = \frac{2GM}{c^2} = 2\left(\frac{M}{m_P}\right)\ell_P. \tag{2.10}$$

The divergence of $Z_g(r)$ as $r \to r_s$ reflects complete asymptotic throttling: local phase–action evolution continues in proper time, but its outward transport rate relative to infinity approaches zero. In QLM this divergence represents an infinite asymptotic transport impedance rather than a breakdown of the local lattice structure.

Equations (2.8)–(2.10) reproduce the standard Schwarzschild time–dilation relations exactly, while admitting a direct interpretation in terms of asymptotic phase–action transport on the Planck lattice.

3 Planck Energy Density Cap (Action-Limited)

3.1 Per–Cell Action Transport

Each Planck four–volume,

$$\mathcal{V}_4 = \ell_P^3 t_P, \tag{3.1}$$

supports exactly one radian of coherent phase evolution and therefore transports one quantum of reduced action,

$$\Delta S = \hbar \qquad \text{per proper-time tick } t_P. \tag{3.2}$$

Because energy is action per unit proper time,

$$E = \hbar\frac{d\theta}{d\tau}, \tag{3.3}$$

this immediately defines the fundamental per–radian Planck energy,

$$E_P = \frac{\hbar}{t_P}. \tag{3.4}$$

3.2 Planck Energy Density

The corresponding Planck energy density is

$$\boxed{u_P \equiv \frac{E_P}{\ell_P^3} = \frac{\hbar}{\ell_P^3 t_P}.} \tag{3.5}$$

This expression follows directly from the covariant phase–flow primitive and the Planck four–volume $\mathcal{V}_4$. It represents the maximal coherent phase–action throughput per unit spatial volume permitted by the lattice.

No additional physical assumptions or phenomenological cutoffs are introduced.

3.3 Local Energy Density Cap

The Quantum Lattice Model therefore enforces the universal local constraint

$$\boxed{u(r) \leq u_P,} \tag{3.6}$$

where $u(r)$ denotes the local proper energy density.

It is convenient to define the dimensionless occupancy fraction

$$\eta(r) \equiv \frac{u(r)}{u_P}, \qquad 0 \leq \eta \leq 1. \tag{3.7}$$

Saturation corresponds to $\eta = 1$, at which the lattice four–volume cell is fully occupied by coherent phase–action transport and admits no additional throughput.

This bound is an intrinsic consequence of the lattice dynamics and introduces no new density scale beyond the QLM primitives.

4 Gravitational Throttling of Energy Transport

Gravitational time dilation relates the local proper time of a lattice cell to the time coordinate measured by an observer at infinity according to

$$Z_g(r) \equiv \frac{dt_\infty}{d\tau} = \left(1 - \frac{r_s}{r}\right)^{-1/2}, \tag{4.1}$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius.

Local phase–action evolution remains governed by the covariant primitive

$$E = \hbar \frac{d\theta}{d\tau}, \tag{4.2}$$

so all local energy densities are defined with respect to proper time and proper volume.

Let Q denote any physical quantity defined locally on the lattice. Rates of change with respect to asymptotic time are related to local rates by the chain rule,

$$\frac{dQ}{dt_\infty} = \frac{d\tau}{dt_\infty}\frac{dQ}{d\tau} = \frac{1}{Z_g(r)}\frac{dQ}{d\tau}. \tag{4.3}$$

Thus, while local lattice update rates remain fixed by the Planck-scale action throughput, any transport rate expressed per unit asymptotic time is throttled by the gravitational impedance factor $Z_g(r)$.

In particular, the local Planck energy density cap

$$u(r) \leq u_P, \tag{4.4}$$

remains invariant in proper units. However, the rate at which energy can be communicated to an observer at infinity is progressively reduced according to

$$\frac{dE}{dt_\infty} = \frac{1}{Z_g(r)}\frac{dE}{d\tau}. \tag{4.5}$$

Gravity therefore acts as a bandwidth limiter on outward energy transport: the local density cap remains fixed at u_P, while the asymptotic transport rate approaches zero as $r \to r_s$.

No modification of the local density bound is required, and the exterior Schwarzschild geometry remains exact.

5 Minimal Interior Completion: Saturated Core

5.1 Maximum Lattice Mass Density

Using the relation $u = \rho c^2$ together with the lattice kinematic identity $c = \ell_P/t_P$, the Planck energy density cap implies a maximum lattice mass density,

$$\boxed{\rho_{\max} = \frac{u_P}{c^2} = \frac{m_P}{\ell_P^3}.} \tag{5.1}$$

This quantity represents the maximum mass that can be supported per Planck cell volume without violating the action–throughput constraint of the lattice.

5.2 Equivalent Action and Mass Forms

Substituting the QLM Planck mass,

$$m_P = \frac{\hbar t_P}{\ell_P^2}, \tag{5.2}$$

into Eq. (5.1) yields

$$\rho_{\max} = \frac{\hbar t_P}{\ell_P^5}. \tag{5.3}$$

An equivalent result follows directly from the Planck energy density,

$$\rho_{\max} = \frac{u_P}{c^2} = \frac{\hbar}{\ell_P^3 t_P}\left(\frac{t_P}{\ell_P}\right)^2 = \frac{\hbar t_P}{\ell_P^5}. \tag{5.4}$$

Thus the action–based, energy–based, and mass–based formulations all reduce to the same invariant lattice density, fixed entirely by the QLM primitives.

5.3 Interior Density Profile

Define the dimensionless occupancy fraction

$$\eta(r) \equiv \frac{u(r)}{u_P}, \qquad 0 \le \eta \le 1. \tag{5.5}$$

Saturation corresponds to $\eta = 1$, at which the lattice four–volume cell is fully occupied by coherent phase–action transport and admits no additional throughput.

Once $\eta = 1$ is reached, no additional coherent phase–action throughput is available. Further compression is therefore dynamically inadmissible within the lattice dynamics.

The Quantum Lattice Model therefore enforces a saturated interior configuration,

$$\rho(r) = \rho_{\max}, \qquad (r \leq R_c), \tag{5.6}$$

where R_c denotes the saturated–core radius defined by the condition $M(R_c) = M$, i.e. the radius at which the enclosed mass equals the total gravitational mass of the object.

This saturated region represents the minimal non–singular interior completion of gravitational collapse.

5.4 Interior Mass Function

For a spherically symmetric saturated core, the enclosed mass for $r \leq R_c$ is

$$M(r) = \frac{4\pi}{3}\,\rho_{\max}\,r^3. \tag{5.7}$$

Imposing the condition $M(R_c) = M$ fixes the core radius uniquely,

$$\boxed{R_c(M) = \left(\frac{3}{4\pi}\frac{M}{m_P}\right)^{1/3}\ell_P.} \tag{5.8}$$

The saturated–core radius therefore scales as $M^{1/3}$, in contrast to the linear scaling of the Schwarzschild radius.

5.5 Exterior Matching

For radii exterior to the saturated core,

$$M(r) = M, \qquad (r \geq R_c), \tag{5.9}$$

and by spherical symmetry (Birkhoff's theorem [5, 7]) the exterior gravitational field is exactly Schwarzschild,

$$r_s = 2\left(\frac{M}{m_P}\right)\ell_P. \tag{5.10}$$

No modification of the exterior geometry is required. The saturated–core completion alters only the interior structure, while all standard exterior predictions of General Relativity remain intact.

The Schwarzschild radius is

$$r_s(M) = \frac{2GM}{c^2} = 2\left(\frac{M}{m_P}\right)\ell_P, \tag{5.11}$$

where the second equality follows from the QLM identities $G = \ell_P^5/(t_P^3\hbar)$, $c = \ell_P/t_P$, and $m_P = \hbar t_P/\ell_P^2$.

The saturated–core radius is

$$R_c(M) = \left(\frac{3}{4\pi}\frac{M}{m_P}\right)^{1/3}\ell_P. \tag{5.12}$$

5.6 Earth–Mass Example

For a mass equal to that of the Earth,

$$M = M_\oplus = 5.9722 \times 10^{24} \text{ kg}, \tag{5.13}$$

the Schwarzschild radius is

$$r_s(M_\oplus) = 8.87 \times 10^{-3} \text{ m}, \tag{5.14}$$

while the corresponding saturated–core radius is

$$R_c(M_\oplus) = 6.50 \times 10^{-25} \text{ m}. \tag{5.15}$$

The ratio of the core radius to the Schwarzschild radius is therefore

$$\frac{R_c}{r_s} = 7.3 \times 10^{-23}. \tag{5.16}$$

5.7 Solar–Mass Example

For a solar–mass object,

$$M = M_\odot = 1.98847 \times 10^{30} \text{ kg}, \tag{5.17}$$

the Schwarzschild radius is

$$r_s(M_\odot) = 2.95 \times 10^{3} \text{ m}, \tag{5.18}$$

while the saturated–core radius is

$$R_c(M_\odot) = 4.49 \times 10^{-23} \text{ m}. \tag{5.19}$$

This yields

$$\frac{R_c}{r_s} = 1.53 \times 10^{-26}. \tag{5.20}$$

5.8 Supermassive Example ($10^6 M_\odot$)

For a supermassive black hole of mass

$$M = 10^6 M_\odot = 1.98847 \times 10^{36} \text{ kg}, \tag{5.21}$$

the Schwarzschild radius is

$$r_s(10^6 M_\odot) = 2.95 \times 10^{9} \text{ m}, \tag{5.22}$$

while the saturated–core radius is

$$R_c(10^6 M_\odot) = 4.49 \times 10^{-21} \text{ m}. \tag{5.23}$$

The ratio becomes

$$\frac{R_c}{r_s} = 1.53 \times 10^{-30}. \tag{5.24}$$

5.9 Analytic Scaling of the Core–to–Horizon Ratio

Using the QLM forms of the Schwarzschild radius and saturated–core radius,

$$r_s(M) = 2\left(\frac{M}{m_P}\right)\ell_P, \tag{5.25}$$

$$R_c(M) = \left(\frac{3}{4\pi}\frac{M}{m_P}\right)^{1/3}\ell_P, \tag{5.26}$$

their ratio is

$$\begin{aligned}\frac{R_c}{r_s} &= \frac{\left(\frac{3}{4\pi}\frac{M}{m_P}\right)^{1/3}\ell_P}{2\left(\frac{M}{m_P}\right)\ell_P} \\ &= \frac{1}{2}\left(\frac{3}{4\pi}\right)^{1/3}\left(\frac{M}{m_P}\right)^{1/3-1}.\end{aligned} \tag{5.27}$$

Therefore,

$$\boxed{\frac{R_c}{r_s} = \left(\frac{3}{32\pi}\right)^{1/3}\left(\frac{m_P}{M}\right)^{2/3}.} \tag{5.28}$$

Thus,

$$\boxed{\frac{R_c}{r_s} \propto M^{-2/3}.} \tag{5.29}$$

Because $r_s \propto M$ while $R_c \propto M^{1/3}$, their ratio decreases as $M^{-2/3}$, explaining the dramatic separation between the saturated–core scale and the horizon scale for astrophysical masses.

5.10 Scaling Confirmation

These examples confirm the analytic scaling relations derived earlier. Because the Schwarzschild radius depends linearly on mass,

$$r_s \propto M, \tag{5.30}$$

while the saturated–core radius scales as

$$R_c \propto M^{1/3}, \tag{5.31}$$

their ratio obeys

$$\frac{R_c}{r_s} \propto M^{-2/3}. \tag{5.32}$$

An increase in mass by a factor of 10^6 therefore increases the Schwarzschild radius by a factor of 10^6, while the saturated–core radius increases only by a factor of

$$(10^6)^{1/3} = 10^2. \tag{5.33}$$

Consequently, the ratio R_c/r_s decreases by a factor of

$$(10^6)^{-2/3} = 10^{-4}, \tag{5.34}$$

consistent with the numerical results obtained above.

The numerical examples therefore verify the fundamental QLM scaling relations and demonstrate that the saturated–core completion becomes progressively more deeply hidden relative to the horizon scale as the total mass increases.

6 QLM Interpretation of Black-Hole Thermodynamics

Classical black-hole thermodynamics assigns a temperature and entropy to Schwarzschild black holes, despite their ostensibly vacuum exterior. In standard treatments, these quantities arise from semiclassical quantum-field theory on curved spacetime.

In contrast, the Quantum Lattice Model (QLM) reproduces the same mass-scaling relations associated with black-hole thermodynamics, but reinterprets their physical meaning in terms of phase–action transport and gravitational impedance.

6.1 Horizon Size as Impedance Divergence

The Schwarzschild radius is

$$r_s = 2\left(\frac{M}{m_P}\right)\ell_P, \tag{6.1}$$

which coincides with the radius at which the gravitational impedance factor

$$Z_g(r) = \left(1 - \frac{r_s}{r}\right)^{-1/2}, \tag{6.2}$$

diverges.

By definition,

$$Z_g(r) = \frac{dt_\infty}{dt_{\text{local}}}, \tag{6.3}$$

so that rates measured relative to asymptotic time are related to local lattice rates by

$$\frac{dQ}{dt_\infty} = \frac{1}{Z_g(r)}\frac{dQ}{dt_{\text{local}}}. \tag{6.4}$$

As $r \to r_s$, one has

$$Z_g(r) \to \infty, \tag{6.5}$$

implying

$$\frac{dQ}{dt_\infty} \to 0. \tag{6.6}$$

Thus, from the perspective of an observer at infinity, outward phase–action transport becomes fully throttled at r_s.

In QLM, the event horizon therefore represents a *bandwidth limit* on phase–action communication to infinity. It is not a physical membrane or material surface, but a limiting impedance condition arising from the lattice transport structure.

The exterior Schwarzschild geometry remains exact; the reinterpretation concerns the transport meaning of the horizon rather than its geometric form.

6.2 Hawking Temperature from Action Throughput

The standard Hawking temperature is [9]

$$T_H = \frac{\hbar c^3}{8\pi G k_B M}, \tag{6.7}$$

which scales inversely with mass.

Using the QLM identities

$$c = \frac{\ell_P}{t_P}, \qquad G = \frac{\ell_P^5}{t_P^3 \hbar}, \tag{6.8}$$

one obtains

$$\frac{G}{c^2} = \frac{\ell_P^3}{\hbar t_P}, \tag{6.9}$$

and therefore

$$\boxed{T_H = \frac{1}{8\pi k_B} \frac{\hbar}{t_P} \frac{1}{M/m_P}.} \tag{6.10}$$

The factor $\hbar/t_P$ represents the fundamental per–tick Planck action energy of the lattice, while the inverse factor $(M/m_P)^{-1}$ reflects the scaling of gravitational impedance with total mass.

In the QLM interpretation, Hawking temperature characterizes the Planck-scale action throughput redshifted by the gravitational impedance associated with a mass M. The inverse mass scaling arises directly from the same impedance structure that defines the Schwarzschild radius.

6.3 Entropy as Frozen Lattice Capacity

The Bekenstein–Hawking entropy is [6, 9]

$$S_{\mathrm{BH}} = \frac{k_B c^3 A}{4\hbar G}, \tag{6.11}$$

with horizon area $A = 4\pi r_s^2$.

Using the QLM form of the Schwarzschild radius,

$$r_s = 2\left(\frac{M}{m_P}\right)\ell_P, \tag{6.12}$$

together with $m_P^2 = \hbar c/G$, one obtains

$$\boxed{S_{\mathrm{BH}} = 4\pi k_B \left(\frac{M}{m_P}\right)^2.} \tag{6.13}$$

Entropy is therefore proportional to the square of the dimensionless mass ratio M/m_P, and equivalently proportional to the horizon area in Planck units.

In the Quantum Lattice Model, this quantity characterizes the total Planck-scale phase–action capacity that becomes asymptotically inaccessible as the gravitational impedance approaches divergence at $r = r_s$.

Entropy is thus not interpreted as counting interior microscopic states. Instead, it measures the effective lattice action capacity associated with the impedance boundary defined by the Schwarzschild radius.

6.4 Unified QLM Scaling

All Schwarzschild black-hole properties are controlled by the single dimensionless mass ratio

$$\frac{M}{m_P}. \tag{6.14}$$

They scale as

$$r_s \propto \frac{M}{m_P}, \tag{6.15}$$

$$T_H \propto \left(\frac{M}{m_P}\right)^{-1}, \tag{6.16}$$

$$S_{\mathrm{BH}} \propto \left(\frac{M}{m_P}\right)^{2}. \tag{6.17}$$

Within the Quantum Lattice Model, these scaling relations emerge naturally once gravitational impedance is expressed in Planck lattice variables. Black-hole thermodynamic behavior is therefore not introduced as an independent principle, but arises from the same mass-dependent impedance structure that determines the Schwarzschild radius.

7 Derived Near–Horizon Implications of the Planck Energy Density Cap

The Planck energy density cap introduced in Secs. 3–5 regularizes the interior of a Schwarzschild black hole by enforcing a finite saturated core. The exterior Schwarzschild geometry, and therefore the gravitational impedance $Z_g(r) = (1 - r_s/r)^{-1/2}$, remains unchanged.

As $r \to r_s$, the impedance diverges, independent of the interior completion. When combined with the action-throughput bound imposed by lattice saturation, this divergence implies that the near-horizon region corresponds to an extreme bandwidth gradient for phase–action transport.

Two potential dynamical consequences may follow from this structure:

(i) inward-propagating gravitational perturbations may encounter an effective high-impedance region once the required local update rate approaches the lattice throughput limit;

(ii) outward action transport near the horizon becomes increasingly bandwidth-limited as measured in the asymptotic frame.

These effects arise from the combined structure of Schwarzschild impedance and the Planck-scale action cap. No modification of the exterior geometry is introduced.

7.1 Impedance Structure and Gravitational Perturbations

In classical General Relativity, the interior of a Schwarzschild black hole acts as a perfect absorber of inward-propagating perturbations.

In the Quantum Lattice Model, however, the existence of a finite saturated core introduces a region in which additional phase–action transport is dynamically inadmissible. If perturbations require local update rates exceeding the Planck throughput bound, the saturated region may behave analogously to a high-impedance termination.

When combined with the exterior Schwarzschild perturbation potential, such an impedance structure could, in principle, produce delayed secondary features in the ringdown signal. A detailed perturbative analysis would be required to quantify the magnitude and observational signature of this effect.

7.2 Radial Propagation and Echo Delay

Impedance perturbations propagate according to the causal lattice wave equation

$$\frac{\partial^2(\delta Z)}{\partial t^2} = c^2 \nabla^2(\delta Z), \tag{7.1}$$

which governs phase–action transport at the universal lattice speed $c = \ell_P/t_P$.

Tortoise Coordinate. To analyze radial propagation in a Schwarzschild exterior, it is convenient to introduce the Regge–Wheeler tortoise coordinate r_* [4], defined so that radial null characteristics propagate at coordinate speed c in the (t, r_*) variables. It is given by

$$r_* \equiv r + r_s \ln\left(\frac{r}{r_s} - 1\right), \qquad (r > r_s), \tag{7.2}$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius.

This coordinate transformation stretches the near–horizon region, mapping $r \to r_s^+$ to $r_* \to -\infty$. It introduces no new physical length scale and is purely geometric.

Exterior Perturbation Barrier. Linearized gravitational perturbations in a Schwarzschild exterior experience an effective Regge–Wheeler potential barrier. The dominant barrier peak occurs near the photon sphere at

$$r_{\text{peak}} \approx \frac{3}{2}\, r_s. \tag{7.3}$$

Evaluating the tortoise coordinate at this location yields

$$r_*(r_{\text{peak}}) = r_{\text{peak}} + r_s \ln\left(\frac{r_{\text{peak}}}{r_s} - 1\right) = \left(\frac{3}{2} - \ln 2\right) r_s \approx 0.8069\, r_s. \tag{7.4}$$

Near–Horizon Reflection Radius. Suppose inward propagation encounters an effective reflection radius r_{ref} located a small coordinate offset above the Schwarzschild radius. We define the offset

$$\delta r \equiv r_{\text{ref}} - r_s, \qquad 0 < \delta r \ll r_s, \tag{7.5}$$

so that

$$r_{\rm ref} = r_s + \delta r. \tag{7.6}$$

Evaluating the tortoise coordinate at $r_{\rm ref}$ gives

$$r_*(r_{\rm ref}) = r_s + \delta r + r_s \ln\left(\frac{\delta r}{r_s}\right). \tag{7.7}$$

For $\delta r \ll r_s$, the logarithmic term dominates, since $\ln(\delta r/r_s) \ll -1$, and we obtain

$$r_*(r_{\rm ref}) \approx r_s \ln\left(\frac{\delta r}{r_s}\right). \tag{7.8}$$

Echo Delay. The round–trip echo delay corresponds to twice the difference in tortoise coordinate between the exterior barrier and the reflection radius,

$$\Delta t_{\rm echo} = \frac{2}{c}\left|r_*(r_{\rm peak}) - r_*(r_{\rm ref})\right|. \tag{7.9}$$

Substituting Eqs. (7.4) and (7.8) gives

$$\Delta t_{\rm echo} \approx \frac{2}{c}\left[0.8069\, r_s - r_s \ln\left(\frac{\delta r}{r_s}\right)\right]. \tag{7.10}$$

For $\delta r \ll r_s$, the logarithmic term dominates over the constant term, yielding the leading behavior

$$\boxed{\Delta t_{\rm echo} \approx \frac{2r_s}{c} \ln\left(\frac{r_s}{\delta r}\right).} \tag{7.11}$$

Interpretation. Even if the effective reflection radius lies arbitrarily close to the Schwarzschild radius, the logarithmic stretching of the tortoise coordinate produces a parametrically large echo delay. This enhancement arises entirely from the exterior Schwarzschild geometry and does not require modification of the metric outside the horizon.

Within the Quantum Lattice Model, such a reflection radius may arise when inward propagation demands a local phase–update rate that approaches the Planck lattice throughput limit. The echo delay therefore follows from the combined structure of Schwarzschild impedance and the Planck-scale action cap.

7.3 Cap–Limited Bandwidth and Determination of the Reflection Radius

A disturbance of asymptotic *angular* frequency ω_∞ experiences gravitational blueshift according to the impedance throttle factor,

$$\omega_{\rm local}(r) = Z_g(r)\,\omega_\infty, \tag{7.12}$$

where $\omega_\infty \equiv d\theta/dt_\infty$ is the angular frequency measured by an asymptotic observer. The gravitational impedance factor is

$$Z_g(r) = \left(1 - \frac{r_s}{r}\right)^{-1/2}, \tag{7.13}$$

with $r_s = 2GM/c^2$ the Schwarzschild radius.

The Planck lattice admits a maximum local update rate fixed by the primitive one–radian–per–tick rule. The corresponding *per–radian* Planck angular frequency is

$$\omega_P \equiv \frac{1}{t_P}, \tag{7.14}$$

where t_P is the fundamental lattice tick.

Inward propagation of that mode ceases once the locally required update rate equals this maximum,

$$\omega_{\text{local}}(r_{\text{ref}}) = \omega_P. \tag{7.15}$$

This condition does not represent an imposed boundary or a physical surface. Rather, it marks the radius r_{ref} at which further inward transport would require super–Planckian phase–update bandwidth and is therefore dynamically forbidden by the lattice throughput constraint.

From Eqs. (7.12)–(7.15) we obtain

$$Z_g(r_{\text{ref}}) = \frac{\omega_P}{\omega_\infty} = \frac{1}{\omega_\infty t_P}, \tag{7.16}$$

and therefore

$$1 - \frac{r_s}{r_{\text{ref}}} = (\omega_\infty t_P)^2. \tag{7.17}$$

Solving Eq. (7.17) gives

$$r_{\text{ref}} = \frac{r_s}{1 - (\omega_\infty t_P)^2}, \tag{7.18}$$

so that the near–horizon offset

$$\delta r \equiv r_{\text{ref}} - r_s = \frac{r_s(\omega_\infty t_P)^2}{1 - (\omega_\infty t_P)^2} \approx r_s(\omega_\infty t_P)^2, \tag{7.19}$$

where the approximation holds for $\omega_\infty t_P \ll 1$.

Using the tortoise coordinate definition (7.2), one finds in the same small–$\omega_\infty t_P$ limit that

$$r_*(r_{\text{ref}}) = r_{\text{ref}} + r_s \ln\left(\frac{r_{\text{ref}}}{r_s} - 1\right) \approx r_s + 2r_s \ln(\omega_\infty t_P), \tag{7.20}$$

while the exterior barrier location gives

$$r_*(r_{\text{peak}}) = \left(\frac{3}{2} - \ln 2\right) r_s \approx 0.8069\, r_s. \tag{7.21}$$

Substitution into the round–trip expression (7.9) yields the echo delay

$$\boxed{\Delta t_{\text{echo}} \approx \frac{2r_s}{c}\left[2\ln\left(\frac{1}{\omega_\infty t_P}\right) - 0.1931\right],} \tag{7.22}$$

valid for $\omega_\infty t_P \ll 1$, where the constant term $0.1931 = 1 - 0.8069$ arises from the difference between $r_*(r_{\text{peak}})$ and the leading r_s contribution to $r_*(r_{\text{ref}})$.

7.4 Frequency–Selective Leakage and Evaporation Endpoint

The same action–throughput constraint governs outward propagation. Consistency with the Planck energy density cap requires that escaping modes satisfy

$$\omega_\infty \leq \frac{\omega_P}{Z_g(r)}, \tag{7.23}$$

where

$$\omega_P \equiv \frac{1}{t_P}, \tag{7.24}$$

and

$$Z_g(r) = \left(1 - \frac{r_s}{r}\right)^{-1/2}. \tag{7.25}$$

As $r \to r_s$, the impedance factor Z_g diverges and the allowed escaping angular frequency range collapses toward zero. Near–horizon leakage is therefore intrinsically low–frequency, while higher–frequency emission can originate only at larger radii where the lattice bandwidth is less severely throttled.

From Sec. 5, gravitational collapse terminates in a finite saturated core of radius

$$R_c(M) = \left(\frac{3}{4\pi}\frac{M}{m_P}\right)^{1/3} \ell_P, \tag{7.26}$$

where m_P is the QLM Planck mass.

Requiring the saturated core to comprise at least one Planck cell in radius imposes the minimal geometric condition

$$\boxed{R_c \geq \ell_P,} \tag{7.27}$$

ensuring that the interior completion remains physically meaningful within the lattice discretization.

Substituting Eq. (7.26) into Eq. (7.27) yields the corresponding minimum mass

$$\boxed{M_{\min} = \frac{4\pi}{3}\, m_P,} \tag{7.28}$$

which represents the smallest mass capable of supporting a saturated Planck-scale core under the QLM density bound.

As $M \to M_{\min}$, further mass loss would require either super–Planckian local update rates or violation of the lattice energy-density cap u_P, both of which are forbidden by the primitive QLM throughput constraint. The semiclassical scaling $T_H \propto M^{-1}$ therefore cannot continue indefinitely. Instead, the evaporation process must break down as $M \to M_{\min}$, implying either evaporation stall or the formation of a long–lived Planck-scale remnant determined entirely by the lattice primitives.

Unified Interpretation

Gravitational–wave echoes and the evaporation endpoint arise from the combination of two structural features: the divergence of gravitational impedance in the Schwarzschild exterior,

and the Planck-scale action–throughput limit enforced by the lattice. Impedance divergence stretches radial propagation in the asymptotic frame, while the density cap limits the allowable local update rate. Together, these mechanisms constrain both inward and outward transport.

Inward perturbations encounter an effective high–impedance region once the required local bandwidth approaches the Planck throughput bound. Outward leakage is progressively filtered to ultra–low–frequency modes and ultimately terminated as the minimum mass scale is approached. Both effects follow from the QLM action–throughput constraint combined with the standard Schwarzschild geometry. No additional postulates, exotic matter, or modification of the exterior gravitational field are introduced.

- The Quantum Lattice Model enforces a universal Planck energy density cap,

$$u_P = \frac{\hbar}{\ell_P^3 t_P}, \tag{7.29}$$

  arising directly from the transport of one quantum of reduced action per lattice four–volume per lattice tick t_P.

- This cap implies a maximum lattice mass density admitting equivalent action–based, energy–based, and mass–based formulations.

- Gravitational time dilation acts as an impedance that throttles energy and action transport without modifying the local lattice bound.

- Gravitational collapse halts in a finite saturated core of radius $R_c(M) \propto M^{1/3}$, providing a minimal non–singular interior completion.

- The exterior gravitational field remains exactly Schwarzschild; no exotic matter, additional parameters, or modifications of General Relativity are introduced.

- Near–horizon consequences of the density cap, combined with impedance divergence, include possible partial reflection of gravitational perturbations (producing delayed echo signatures) and intrinsic frequency filtering of outward action leakage, yielding a natural evaporation endpoint.

- Black–hole temperature and entropy arise as *derived, emergent* measures of redshifted and frozen lattice action capacity.

The propagating impedance perturbations responsible for these near-horizon effects also govern gravitational-wave radiation in the QLM; a detailed lattice-level derivation is provided in Appendix B.

8 Gravitational Waves as Vacuum–Impedance Collapse

This appendix provides a derivation of the dynamical impedance perturbations used in the main text. Within the Quantum Lattice Model (QLM), gravitational radiation is interpreted as propagating modulations of lattice phase–flow impedance.

This reinterpretation does not modify the exterior Schwarzschild solutions employed in the main body of the paper. The resulting wave equations reproduce the same propagation speed, causal structure, and asymptotic behavior as standard linearized General Relativity.

In QLM, the metric perturbation degrees of freedom of classical gravity correspond to fluctuations in the local impedance governing phase–action transport. The geometric description of curvature is therefore retained at the level of observable predictions, while the underlying dynamical variables are expressed in lattice transport form.

8.1 Impedance Decomposition

In the Quantum Lattice Model, gravity is interpreted as a modulation of lattice phase–flow bandwidth. We decompose the total lattice impedance as

$$Z(\mathbf{x}, t) = Z_0 + Z_g(\mathbf{x}) + \delta Z(\mathbf{x}, t), \tag{8.1}$$

where Z_0 is the vacuum impedance corresponding to minimal phase–volume, $Z_g(\mathbf{x})$ is the static gravitational impedance sourced by a stationary mass distribution, and $\delta Z(\mathbf{x}, t)$ represents time–dependent impedance perturbations.

Only $\delta Z(\mathbf{x}, t)$ contributes to gravitational radiation. A static mass distribution produces no propagating disturbance, consistent with the absence of monopole radiation in standard gravity.

8.2 Action Conservation and Causality

In the Quantum Lattice Model (QLM), energy is not treated as a primitive substance but as action per unit proper time, $E = \hbar \frac{d\theta}{d\tau}$. The fundamental conserved quantity is therefore reduced action density.

Let $\rho_S(\mathbf{x}, t)$ denote the local action density (reduced action per unit spatial volume), and $\mathbf{J}_S(\mathbf{x}, t)$ the associated action–flux density, representing transport of reduced action per unit area per unit time.

In the continuum limit of lattice transport, conservation of reduced action takes the form of a local continuity equation,

$$\frac{\partial \rho_S}{\partial t} + \nabla \cdot \mathbf{J}_S = 0, \tag{8.2}$$

which expresses the statement that reduced action cannot be created or destroyed within a lattice region.

Each Planck cell of spatial volume ℓ_P^3 transports at most one quantum of reduced action $\hbar$ per lattice tick t_P. The maximal local action density is therefore

$$\rho_{S,\max} = \frac{\hbar}{\ell_P^3}. \tag{8.3}$$

Because lattice updates occur only between neighboring cells and advance by one cell per tick, no disturbance may propagate faster than the primitive lattice speed

$$c = \frac{\ell_P}{t_P}. \tag{8.4}$$

Therefore, time–dependent impedance variations cannot remain localized and must propagate outward as a causal lattice response, with propagation speed bounded by c.

8.3 Vacuum Relaxation as the Restoring Mechanism

When a moving mass transiently modifies the surrounding lattice, the local impedance increases relative to its vacuum value. This configuration is sustained only while the mass distribution maintains the corresponding phase–flow gradient.

If the mass shifts, the impedance field becomes dynamically inconsistent with the new source configuration. Because the action density is bounded by the primitive lattice scale $\rho_{S,\max} = \hbar/\ell_P^3$ and may only be redistributed by nearest–neighbor updates at speed $c = \ell_P/t_P$, the lattice must readjust through causal transport in order to restore a self-consistent phase–flow distribution.

The minimal impedance state corresponds to uniform phase transport. Deviations from this state require sustained action throughput gradients. When those gradients change in time, the impedance field relaxes through causal propagation governed by the action continuity relation (8.2) and the finite update speed of the lattice.

Thus:

- impedance increase corresponds to lattice phase–flow displacement,
- continuity of reduced action enforces redistribution,
- finite lattice update speed enforces causal propagation,
- outward-propagating impedance perturbations arise as the minimal relaxation channel consistent with these constraints.

Gravitational radiation therefore arises from time-dependent reconfiguration of the impedance field. Energy carried by the radiation corresponds to transported reduced action associated with these propagating impedance disturbances.

8.4 Impedance Wave Equation

Perturbations of the local reduced action density arise from corresponding variations in lattice impedance. Because no dimensional scale exists beyond the primitive lattice quantities $\{\hbar, \ell_P, t_P\}$, small deviations in action density must be proportional to the maximal lattice density scale,

$$\delta\rho_S = \frac{\hbar}{\ell_P^3}\,\delta Z. \tag{8.5}$$

Starting from the action continuity relation

$$\frac{\partial\rho_S}{\partial t} + \nabla\cdot\mathbf{J}_S = 0, \tag{8.6}$$

and linearizing about the vacuum state gives

$$\frac{\hbar}{\ell_P^3}\frac{\partial(\delta Z)}{\partial t} + \nabla\cdot\mathbf{J}_S = 0. \tag{8.7}$$

Because redistribution of reduced action occurs only through nearest–neighbor updates and advances by one lattice spacing ℓ_P per tick t_P, the action flux must respond to spatial gradients at the finite lattice speed

$$c = \frac{\ell_P}{t_P}. \tag{8.8}$$

Dimensional closure under the primitive lattice scales uniquely fixes the causal flux response to be

$$\frac{\partial \mathbf{J}_S}{\partial t} = -\frac{\hbar}{\ell_P^3} c^2 \, \nabla(\delta Z). \tag{8.9}$$

Taking a further time derivative of Eq. (8.7) and substituting Eq. (8.9) yields

$$\frac{\partial^2(\delta Z)}{\partial t^2} = c^2 \, \nabla^2(\delta Z). \tag{8.10}$$

Thus impedance perturbations propagate as causal lattice waves at the universal phase speed c, determined uniquely by the primitive identity $c = \ell_P/t_P$. No phenomenological coefficients or free parameters are introduced.

In the continuum limit, this wave equation reproduces the same propagation structure as linearized general relativity, expressed here in lattice transport variables.

8.5 Maxwell–Like First–Order Formulation

The impedance dynamics may equivalently be expressed in a first–order, Maxwell–like system derived directly from lattice primitives.

Using the primitive relation between action density perturbations and impedance variations,

$$\delta\rho_S = \frac{\hbar}{\ell_P^3} \, \delta Z, \tag{8.11}$$

the action continuity equation becomes

$$\boxed{\frac{\partial}{\partial t}\left(\frac{\hbar}{\ell_P^3} \, \delta Z\right) + \nabla \cdot \mathbf{J}_S = 0} \qquad \text{(action continuity)}. \tag{8.12}$$

Finite lattice update speed enforces the causal flux response

$$\boxed{\frac{\partial \mathbf{J}_S}{\partial t} + \frac{\hbar}{\ell_P^3} \, c^2 \nabla(\delta Z) = 0} \qquad \text{(causal lattice response)}. \tag{8.13}$$

Together, Eqs. (8.12)–(8.13) reproduce the impedance wave equation (8.10) and make explicit the causal transport structure of gravitational radiation in the Quantum Lattice Model. No additional phenomenological coefficients are introduced; the propagation speed is fixed uniquely by the primitive identity $c = \ell_P/t_P$.

8.6 Relation to Observables

Interferometric detectors respond to phase delays accumulated along their optical paths. In the QLM transport formulation, a passing impedance perturbation modifies the effective phase–flow rate along the arm, producing a fractional path change proportional to the line integral of the dimensionless impedance variation.

To leading order in the perturbation amplitude, the measured strain scales as

$$\frac{\Delta L}{L} \propto \int_{\text{arm}} \frac{\delta Z(\mathbf{x}, t)}{Z_0}\, d\lambda, \tag{8.14}$$

where $d\lambda$ denotes the optical path element along the detector arm and Z_0 is the vacuum lattice impedance.

Because $\delta Z/Z_0$ is dimensionless, the integral represents a fractional phase delay accumulated along the arm.

In the continuum limit, this scaling reproduces the same observable strain response predicted by linearized general relativity, expressed here in lattice impedance variables.

8.7 Notation and Symbols

The following symbols are used throughout the paper. All quantities are defined at their first occurrence; this table is provided for reference.

- ℓ_P — Planck lattice cell edge length.
- t_P — Planck lattice tick (fundamental time step).
- $\hbar$ — Reduced quantum of action.
- c — Lattice phase speed, $c = \ell_P/t_P$.
- $\mathcal{V}_P$ — Single Planck cell volume, ℓ_P^3.
- E_P — Per–radian Planck energy, $\hbar/t_P$.
- m_P — QLM Planck mass, $\hbar t_P/\ell_P^2$.
- u_P — Planck energy density cap, $\hbar/(\ell_P^3 t_P)$.
- $\rho_{\max}$ — Maximum lattice mass density, m_P/ℓ_P^3.
- M — Total gravitational mass sourcing the lattice.
- r_s — Schwarzschild radius, $2GM/c^2 = 2(M/m_P)\ell_P$.
- R_c — Saturated–core radius.
- Z_0 — Vacuum lattice impedance.
- $Z_g(r)$ — Gravitational impedance (time–dilation) factor.

- $\rho_S(\mathbf{x}, t)$ — Local reduced action density.
- $\mathbf{J}_S$ — Reduced action flux density.
- δZ — Impedance perturbation.

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